Ex: 12/12/19

Implement a chat program using Socket programming

Aim:

To implement a simple chat client server application using TCP sockets for communication between clients through a server.

ALGORITHM:

* Strat the TCP server and create a serversocket.
* Accept the client connection.
* Create a thread to handle each client.
* Read message sent by the client.
* Broadcast the message to other connected clients.
* continue chatting until /quit is entered.
* Close the connection.

CODE:

ChatServer

```
import java.io.*;
import java.net.*;
import java.util.*;
public class ChatServer {
    static Set<PrintWriter> clients = new HashSet<>();
    public static void main (String[] args) throws Exception{
        ServerSocket server = new ServerSocket (5000);
        System.out.println ("Chat Server Started");
        while (true)
        {
            Socket s= Server.accept();
            new Thread() -> {
```

```
        new InputStreamReader (S.getInputStream()));
    PrintWriter out = new PrintWriter (
      S.getoutputStream(), true);
      clients.add (out);
      String msg;
      While ((msg = in.readLine()) != null) {
        System.out.println (msg);
        for (PrintWriter p: clients)
          p.println (msg);
      }
    }catch (Exception e) {
    }
  }).start();
 }
}
}
```

ChatClient

```
import java.io.*;
import java.net.*;
public class ChatClient {
  public static void main (String[] args) throws Exception {
    Socket s=new Socket ("localhost", 5000);
    PrintWriter out =new PrintWriter(
      S.getOutput Stream (), true);
    BufferedReader kb = new BufferedReader (
      new InputStreamReader (System.in));
    new Thread (() -> {
      try {
        String msg;
        while ((msg =in.readline())!=null)
          system.out.println(msg);
      } catch (Exception e) {
```

OUTPUT :-

Server:

Chat Server Started

Alice joined the chat

Prodhap joined the chat

Alice : Hello

Prodhap : Hi

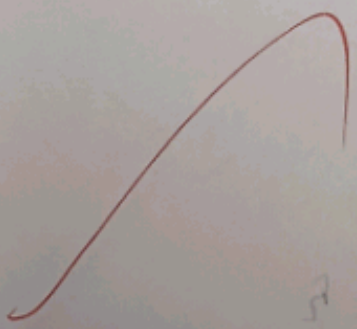
Client :

Enter your name:

Alice

Hello

prodhap : Hi



```
}).start();
String msg;
while ((msg = kb.readLine()) != null) {
    out.println(msg);
    if (msg.equalsIgnoreCase("/quit"))
        break;
}
s.close();
}
}
```

RESULT:

Thus, the Chat Client - Server application was successfully implemented using TCP sockets.